LESSON PLAN

EWIS (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Radar system introduction to radar system, primary and secondary radar system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Radar system introduction to radar system, primary and secondary radar system
5. Specific Objectives	21.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Radar Block Diagrams and Signal Flow; Safety: No live equipment handling; Supervision: Instructor monitors PPT navigation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the fundamental definition of Radar?
(b) List three basic components of a communication system.
(c) Define the term 'Echo' in the context of radar.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic Theory of Communication • Radar Signal Transmission Principles • Wave Propagation Characteristics • Antenna Functionality Overview • Receiver Processing Basics	L & Ds	PPT, IB	15 min
(b)	• Primary Radar System Architecture • Secondary Radar System Architecture • Transponder Operation Principles • Interrogator and Reply Unit Functions • Comparison of Primary and Secondary Systems	L & Ds	PPT, PDF	20 min
(c)	• Radar Display Interpretation • Blind Sector and Coverage Limits • Noise and Clutter Mitigation • System Maintenance Overview • Operational Safety Protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the key difference between Primary and Secondary radar systems.
(b) Describe the role of the transponder in a secondary radar system.
(c) Identify two factors that affect radar range accuracy.

12. Home Assignment and Follow Up	Prepare a brief note on the purpose and technical details of VOR/ILS/MKR systems to be discussed in LP 04.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose and technical details of VOR/ILS/ MKR system	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min